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Chapter 12

Drinks Warehousing Overview



Introduction

Drinks Warehousing provides different levels of support for different organisations, or indeed different warehouses within the same organisation. At the simplest level it may be used to provide a tracking system for a limited range of key items. This can be expanded to encompass all aspects of warehouse control and management.

The map of the warehouse can include proximity algorithms which are used as one of the ways of deciding where items should be located.

Picking and marshalling may either be order based or, depending on the organisational needs of the warehouse, can use the characteristics of journey, customer, and due date. Full account is taken of expiry dates, lots or batches, pallet numbers and warehouse organisation when stock is allocated to picking instructions.

The module is designed to fit into the operational environment of a warehouse, in that the minimum of data entry is required to operate it. Lists of tasks are displayed and all that is normally required is simple confirmation. The structure of the database simplifies the interface to technical advances in Drinks Warehousing such as hand-held terminals.

Relationships to Other Modules

Drinks Warehousing is a fully standardised module. It is one of the group of Advanced Application Modules which extend the functionality of Base Application Modules.

The only module which is a pre-requisite for the operation of Drinks Warehousing is Drinks Inventory Management.

The module, however, is designed to be used in conjunction with, and enhance the following modules:

- Drinks Sales Order Processing.
- Drinks Transport Planning.
- Drinks Purchase Management.
- Production Definition Management.
- Production Order Control.
- Production Schedule Control.

When Drinks Warehousing is installed and a stockroom has been fully activated as a warehouse, links are switched on between the operational modules and the warehouse that provides the additional location management and control. Thus, items received against a purchase order can automatically have pallet labels printed that include put-away instructions. Despatches to be made for sales orders recorded by the Drinks Sales Order Processing module and issues of inputs to production orders and schedules can be picked, marshalled and despatched using the enhanced facilities of Drinks Warehousing. Receipts from production orders and schedules can be made into a warehouse and put-away routines are invoked.

Module Configuration

As with all modules, Drinks Warehousing can be operated for a number of companies, the characteristics of each being maintained on control files. Drinks Warehousing uses the control data stored for Drinks Inventory Management. The precise control in this module, however, is required at the level of the Warehouse because it will be appropriate to have different policies in different parts of the organisation. When the module is installed, a warehouse may be fully configured before it is activated. This enables the proper degree of preparation and checking to take place prior to the start of full operation.

The policies which can be controlled at each warehouse are:

- The module to give instructions or to record actions.
- Check digits to be used for confirmation of actions.
- Stock rotation.
- The sequence in which action lists are produced.

All actions in the operational modules resulting in a Drinks Inventory receipt will cause the item to be automatically located in the receiving location. Similarly, when picking is required, the module will recommend that stock for a number of items from certain locations should be moved to the marshalling location prior to issue. Any locations which have been set up in the warehouse may be defined as these two locations. However, it is very strongly recommended that warehouse areas be set up so that each contains a single location which has a Location Code which is the 'normal' name for receiving and marshalling in the warehouse. These areas can be set up to satisfy the requirements of each individual warehouse.

Accuracy and discipline are the fundamentals of the Drinks Warehousing operation. It is therefore vital that, if unexpected actions are necessary, the supervisor is notified immediately. If errors are detected by the module, a message can be directed instantly to a supervisor or to a terminal if the supervisor is not currently signed on.

Each item which is managed by Drinks Warehousing requires an item / warehouse profile in order to define its rules and characteristics. Defaults can be set up for the warehouse to minimise the amount of data entry required during the maintenance of these profiles.

A user of Drinks Warehousing will normally be at a particular site and therefore it is possible to define a default warehouse. This simplifies the day-to-day operational procedures particularly for recording transactions and making enquiries.

Maintenance of Reference Data

As this is an Advanced Application Module, much of the primary data has already been set up as part of the base modules. There is, however, additional information which must be entered in order to obtain the maximum benefit from the module.

There are a number of common features to aid the maintenance of the data, these are:

- The standard item and descriptions search routines have been adopted.
- The module makes the minimum use of 'codes' and where they are employed a prompt is always available to assist in selecting the correct one.
- Selection is normally from a displayed list.

Information for the following may be maintained:

- Warehouse profiles.
- Area profiles and dimensions.
- Item/warehouse profiles.
- Packaging types.
- Location types.
- Warehouse map.
- Action rules.
- Resources.
- Check digits.
- List profiles.

Warehouse Profile

The warehouse profile is used for five distinct functions. Firstly, to set up the policies for the warehouse, e.g. should check digits be used for confirmation. Secondly, to set up the special areas for receiving and marshalling. Thirdly, to define the target for any error messages produced by the module, such as no location being suggested for the picking of stock based on the rules defined. Fourthly, to define defaults that are used in the item/warehouse profiles. These defaults are set to save time when defining the item/warehouse profiles. Finally, to activate or de-activate the warehouse.

Area Profiles and Dimensions

The purpose of setting up a warehouse is to record and manage the locations and quantities of items stored within that warehouse. Before this can be done it is necessary to define the shape and structure of the warehouse to the module.

The module treats the warehouse as a series of areas, each of which is divided into a number of locations. For example, to reflect the physical nature of a warehouse you could define three areas, one called 'Floor' which is used for the receiving and marshalling locations, another called 'Main Store' which represents

most of the aisles, rows and bays in the warehouse and another called 'Picking' which represents the pick locations.

Each area is defined using up to four dimensions, such as aisle, bay, row, etc.. The dimensions are used to build up the code that identifies individual locations. The code for each dimension may be up to six characters in length, but the final location code (the concatenation of all the dimension codes) may not exceed twelve characters.

Item / Warehouse Profile

Each item to be kept in a warehouse must be associated with a location. If the module is to be used effectively, it will make recommendations regarding where items will be put, when and from where replenishment should take place, and from where items should be picked. The management of an item will vary between warehouses in the same company. It is necessary to define to the module the parameters for the items as held in each warehouse. The information and parameters are held under five headings.

Packaging

This relates the item in inventory stockroom issue units of measure to the pack types used in the warehouse. It also defines the basis of stock rotation.

Fixed Locations

Despite the many advantages of 'random' locations there are some items which are best kept in fixed locations.

Put-Away

This defines the processing steps to be used by the module when making recommendations as to where an item should be put. This includes trying the picking locations, using the item location rule and using the area location rule.

Commit

This defines the processing steps taken by the module when making recommendations of from where an item should be picked. These steps include the consideration of stock rotation, picking location, fixed locations, item and area rules.

Pick Location

Control data for the picking location can be set up including replenishment trigger and the replenishment quantity. It is also possible to define the maximum, in a single pick, to be taken from this location before the pick is made direct from bulk.

Pack Types

Many of the movements in a warehouse are performed using pallets or containers of some kind. These can be defined to the module and then the relationship between the pack and item can be created. A default pack type must be set up for each item to obviate the necessity to enter a pack type for each movement.

Location Types

Depending on the purpose of the warehouse there will be a number of types of locations. The size, shape, and use of these will be chosen to meet particular

storage requirements including the number and distribution of pack types. This information is used by the module, together with the number of packs per location, to decide whether or not a particular receipt should be put away in a location.

When the location types have been established, their distribution throughout the warehouse areas can be defined. For example, there may be standard pallet racking which will accept all standard pallets, picking locations which are equivalent to one and a half pallets and locations on the top of the racks which are used to store rolls of packing material and cardboard boxes.

In addition to these characteristics it is possible to remove locations from use by tagging them as void, and to limit their use to single-item occupancy or to specify multi-item occupancy as being allowed.

Warehouse Map

The warehouse map assigns location types to locations. This determines the characteristics of the locations.

For example, the locations within an area set-up called 'Main Store' can be set to have a location type of 'Standard Pallet Racking' which limits the locations to being capable of storing only one pallet at a time.

Location Rules

When a warehouse movement takes place it should only ever be done so as a result of an instruction. Before the instruction is issued, a decision is made as to which is the 'best' location to be used. The 'best' location will depend upon a number of factors such as:

- The item itself.
- How it is packed.
- Does it require special conditions?
- Is the item picked from a picking location?
- Can mixed 'lots' be put in the same location?
- Must the receiving area be cleared quickly?

The warehouse operator uses considerable experience to 'unconsciously' make such decisions. It is possible to create rules in the module that, to some extent, mimic these decisions. Combinations of the following can be set up to control the recommendations for put-away, pick and replenish.

- Use the pick location.
- Use the fixed locations.
- Use an item rule. An item rule is intended to reproduce the kind of review that would be done by line-of-sight checks. For example, if a replenishment is to be performed to the picking location from the nearest full pallet or, conversely, if a pallet is to be stored as close to the pick location as possible. Creating the rule involves defining a series of locations to be tested in relation to the pick location.
- Use an area rule. This rule is intended to review a large number of locations to answer such questions as "Find the first not-full location in this part of the

warehouse which is closest to the receiving bay”, or “Select randomly a not-full location.” This type of rule may make use of the proximity data set up for an area or the randomising algorithm.

- Use stock rotation dates. These dates include expiry and arrival date.

Obviously different types of moves are conditioned by different factors. For example, stock rotation is used for picking and replenishment but not for putting away.

Resources

Resources relate to the warehouse equipment, such as trolleys and fork-lift trucks. A resource can be defined to the system and then related to pack types within a location type. For example, a resource of ‘trolley’ could be related to a pack type of ‘pallet’ in a location type of ‘floor’; whereas the resource of ‘fork lift truck’ would be related to the pack type of ‘pallet’ if the location type was ‘standard racking’ as the location may not be at ground level.

Check Digits

Check Digits or Return Codes can be used within the module to ensure that the warehouse staff have visited the defined location where the check digit is displayed. The confirmation of the action can then be by the entry of the return code or check digit. In the case of an intra-warehouse movement, both check digits must be entered. Check digit usage is controlled as follows:

- The use of check digits is requested for the warehouse.
- A set of default check digits can be set up for the warehouse.
- This may be overridden for each or for all areas.
- If no defaults are set up but check digits have been requested, modulus 23 is used as a base to generate digits.

Processing

Most of the work in a warehouse can be divided into three types of tasks:

- Putting away items.
- Moving items to optimise storage, picking and despatching.
- Picking and despatching.

One of the main functions of Drinks Warehousing is to make recommendations for each of the three tasks, and then produce instruction lists on paper and on screen. Once these tasks have been completed, they can be confirmed either by the entry of a '1' against the action, or, if check digits are being used, the entry of the appropriate digits. If check digits are not required, actions can be set to be confirmed by default. This option will still allow you to override those actions.

Actions may be changed to indicate that different locations were used when carrying out the instructions. This is necessary to keep the module up to date with what is happening within the warehouse.

In order to manage computer resources, some of the processing can be carried out in a background subsystem or may be run in batches instead of being performed interactively. Running jobs in batches can be especially useful for printing, where special stationery for work planning is required.

Receiving and Putting Away

There are a number of steps in recording the receipt of an item and putting it away within the warehouse. These steps are either the recording of specific actions or the processing and making of recommendations by the module. The recording of all external events is done by using interactive procedures. In order to manage the computer resources, some of the processing is carried out in a "continually processing sub-system". This waits for the next instructions and then carries them out. As a third possibility, work can be performed in small batches. This is particularly true where printing is involved and special stationery for work planning is required. In some instances you are provided with a choice in the way processing is performed. The steps involved with receiving and putting away are as follows:

Drinks Inventory Receipt

This is recorded in one of the operational modules of , i.e. Drinks Purchase Management, Drinks Inventory Management, Production or Drinks Sales Order Processing. These actions are interactive procedures. The result is an increase in the inventory balance at the stockroom. This must be reflected as an increase in the warehouse. The continually processing sub-system moves the item into the location denoted as RECEIVING. The warehouse and inventory are now in balance. No explicit additional action is required in the operational module to achieve the second posting. However, as Drinks Warehousing can operate in multiple packaging types, an opportunity is provided to confirm or change the pack type at the time the receipt is made into a warehouse.

The receiving sub-system performs the following functions:

- Moves the item into the receiving location.
- Determines the number of packs.
- Identifies the rules for putting the item away.
- Makes a sufficient number of recommendations to put the item away into the warehouse.
- One recommendation is produced for each unique combination of Item/lot/pack type and location.

The item is now planned out of receiving and planned into the locations.

Printing Documents

Before an action can be confirmed, an instruction must have been generated by the module. If the item is packed such that pallet labels have been requested, they will be produced; if not a list will be printed. In order to provide the proper control, these are only printed in the warehouse when requested.

Confirmation of the Action

As with all recording of actions, this procedure is interactive. There is an obvious advantage in this recording being as close to real-time processing as possible. Confirmation of the put-away can either be by entry of a single character or by a check digit which can only be found by visiting the location.

In order to guarantee the best possible response times while recording put-away transactions, the update of the database may be moved into a continuously updating sub-system. Update is normally complete within seconds of entry being finished.

Intra-Warehouse Move

Movements inside the warehouse or intra-warehouse movements are performed to improve the organisation of items in order to reduce work at more critical times.

It is necessary not only to be able to record that a move is required, but also that a move has been done. A move may be occasioned by a number of factors.

The most common is letting down from bulk to a picking area. At times it may be decided to refurbish part of the warehouse which may result in the movement of items out of a specific part of the warehouse. Finally, the need for action may be so immediate that the action to relocate is taken and then reported to the module after the event. In a well planned warehouse such events will be rare.

In common with the other two types of actions, putting away and picking, a list of instructions is produced which you can see on a screen and confirm or change.

A distinction is drawn between movements generated by the module for replenishment and those manually requested. These may be produced on separate lists in order to assign the proper priority for the different actions. When confirmation is being performed, different overrides are appropriate for the different moves. For example, if a replenishment of a picking location has been requested by the module, it is not possible to change the item or the 'move to' location because the intention of the instruction is to replenish a fixed picking location.

Picking and Despatching

The initial allocation of stock to sales orders takes place in the Drinks Sales Order Processing or Production Control & Costing modules. This allocation is done against the available stock figure in the nominated stockroom but does not identify in which location the stock will be found to meet that particular requirement. This is performed in Drinks Warehousing. The particular location being decided upon by reviewing such factors as:

- The quantity to despatch.
- Stock rotation.
- Physical proximity of location.
- The practicality of picking a number of order lines together

The picking and despatching cycle has the following events:

- Request Picking Lists.
- Confirm Pick.
- Confirm Despatch or Confirm Issue.

Request Picking Lists

This is performed by Drinks Warehousing and considers requests to despatch generated from the Drinks Sales Order Processing module or requests to issue generated from the Production Control & Costing module. A series of tasks are performed which result in the printing of the documents necessary to pick and despatch the items. The first task sets up the parameters which control the others which are to:

- Select a range or group of orders.
- Sequence and combine them.
- Decide from which locations the stock should be picked.
- Print a pick list organised to suit the warehouse.
- Print appropriate marshalling lists which cross-reference picking lists and despatch or issue notes.
- Print the despatch/issue notes.



Note: Kit parent and non-stock items are not committed and do not print on the warehouse pick list.

Confirm Pick

This is essentially similar to confirmation of all the warehouse actions and can be performed by entering either the check digit or return code. It is possible to override the module instruction by changing both the location and the quantity. This will result in an error being recorded. The result of the confirmation is to transfer the stock to the marshalling location.

Confirm Assembly

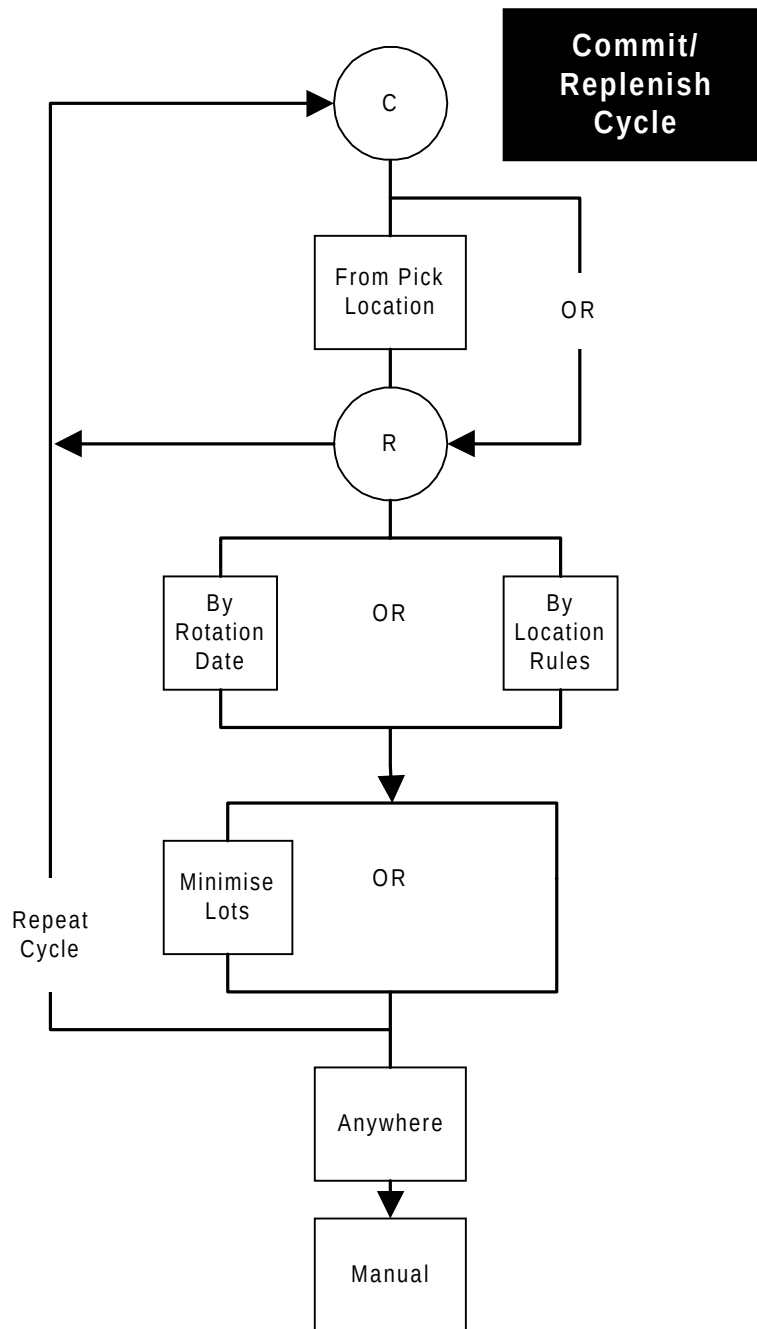
Confirmation of assembly may be performed from the sales marshalling location. This confirmation is carried out at order level using the standard picking note/despatch note as produced from the Drinks Sales Order Processing module.

Reports are available from the module to aid the marshalling of complex loads which link picking lists and despatch/issue notes. If the despatch has taken place as intended, this may be confirmed using a single function key. Provision is made for the addition of delivery and freight charges.

Confirm Issue

From the production marshalling location, confirmation of issue may be performed. This confirmation is carried out at order level using the standard shop documentation as produced from the Production Control & Costing module.

Reports are available from the module to aid the marshalling of complex loads which link picking lists and despatch/issue notes. If the issue has taken place as intended, it may be confirmed using a single function key.



The process is performed for each commit action. A commit action is at the level of Item, lot, Stock rotation date, pack type or pack number.

Enquiries

Enquiries can be grouped into one of three categories.

- Status.
- Action Lists.
- Movements.

More simply, these can be regarded as - Where are things? What is to be done, and, What has happened?

Status

The status enquiries either start from examining what is in a location or range of locations, or in which location an item is stored. Whenever a number of locations are to be displayed it may be appropriate not only to select them but also to sequence them. The most obvious sequence is the Location Code and this will be the one which is normally used; however, if empty locations are being displayed, it may be appropriate to list them by an Area or Item Rule. In this way manual decisions can be made on the same sort of basis as module decisions.

Whichever entry is chosen into the status displays, it is possible to examine:

- Item/Location Details.
- Location Contents.
- Location Details.

Action Lists

There are three types of action lists:

- Put Away.
- Move.
- Pick.

The lists which have not been finished are displayed. The details of despatch instructions sent but not yet actioned are available for enquiry with direct access to the full order enquiry within the Drinks Sales Order Processing module.

Movements

This class of enquiry includes the warehouse movements and the reported error conditions. If an action is requested but not then confirmed as instructed, an error record is written. These can be displayed. Multiple selection criteria are provided for both movements and errors enquiries.

Reports

As part of the general policy of modules selection and sequencing parameters are available before a report is run. In general, standard reports are printed on standard listing paper with 132 print positions and 6 lines to the inch.

All reports are printed with the procedure or program number in the top left, and the page number, date and time to the top right. As with enquiries, the reports can be grouped into Status, Actions and Movements.

Overall Implementation Considerations

It is of vital importance to the success of any software implementation that certain issues are considered before detailed steps are undertaken. In case of Drinks Warehousing these issues are:

- How Drinks Warehousing and other Modules should be used.
- How many warehouses should be set up.

Implementation Procedure

There are two possible implementation procedures that could apply:

- Drinks Warehousing with Drinks Inventory Management already installed.
- Drinks Warehousing with Drinks Inventory Management newly installed.

The appropriate procedure should be used as a guide.

It is strongly advised that whichever procedure is used a project plan is followed in order that the necessary data can be captured and recorded in advance of the entry of the stock count.

The basic steps for a successful implementation are:

- Master file data collection and input.
- Work load organisation.
- Suspend processing of stock.
- Initial date take-on/reconciliation.
- Resume processing of stock.

With Drinks Inventory Management already Installed

It is assumed that users are familiar with Drinks Inventory Management.

Master File Data

The data required to make the Drinks Warehousing Module functional must be defined. The amount of effort required will depend upon the number of items, locations, pack and location type details which need to be defined.

Work Load Organisation

It is essential that during the stock count exercise and the early stages of processing with the Drinks Warehousing module that the movement of stock is controlled. To help achieve this it is advisable to minimise the activities for receiving and issuing stock.

Receipts

Any receipts which have been booked into stock but have not been put away should be reviewed. The possible actions that can be performed for each receipt are:

- Put the items away, thus reducing the items at the receiving area to those not yet booked into stock.
- If the receipts were made through Purchase Management the items could be transferred back to either goods inwards or inspection. The items would then be received into stock after the activation of the warehouse.
- Leave the items at the receiving location. The stock should be recorded at Data Take-on and then moved manually after the activation of the Warehouse.

Issues

All requests for issuing stock, e.g. order despatch, should be processed such that the items to be despatched have been picked and moved to the Marshalling location. Obviously, it would make processing simpler if the items were despatched and the actions confirmed.

Leaving items at the Marshalling locations means that the full details required for Confirmation of Despatch must be entered.

Suspend Processing of Stock

The movement of stock into, around and out of the warehouse must be suspended whilst the stock count for Data Take-on is performed and reconciled.

From this point until the warehouse is activated, the processing of Drinks Inventory, Drinks Purchasing, Production and Drinks Sales Order Processing must be suspended. Outstanding transactions must be completed.

Initial Data Take-On / Reconciliation

A full stock count is required to include every item and every location in the Warehouse. The details are entered using the Data Take-On data entry programs.

The count should be reconciled to the Drinks Inventory lot and stockroom balances, discrepancies are shown on the report. Any variances should be verified. Once the count is cleared as correct, the location details should be updated using the Reconciliation with Update option. This will also make appropriate adjustments to the Drinks Inventory balances.

Resume Processing of Stock

The warehouse must be activated. This is done by taking the **F14** function key from the Drinks Warehouse Profile Maintenance option.

Transaction processing can be resumed in all areas of the warehouse. Transactions can be recorded in the modules previously suspended.

Receipts and issues are now subject to the control of the Drinks Warehousing Module.

With Drinks Inventory Management newly Installed

Appropriate training needs to be undertaken to enable users to understand both the Inventory and Drinks Warehousing Modules.

Master File Data Collection and Input

The data required to make both Drinks Inventory and Drinks Warehousing functional must be defined. Care must be taken to ensure that dependencies are entered in the appropriate order. The amount of effort required will depend upon the number of items, location, pack and location type details that need to be defined.

Work Load Organisation

It is essential that during the stock count exercise and the early stages of processing with the Drinks Warehousing module the movement of stock is controlled.

Receipts

Items that are in the receiving area should be reviewed. The possible actions that can be performed are:

- Put the items away before the stock count.
- Record, at data take-on, the items in the receiving area. Manual moves will be required after the activation of the warehouse.
- Leave the items at the receiving location, do not record the stock during data take-on. The items are received after activation of the warehouse.

Obviously, the status of stock at the receiving location must reflect the status of stock received into Inventory.

Issues

Items to be despatched should be processed through to the marshalling area. Obviously, the stock status must reflect the requests for despatch in other modules.

It is better to start with as few incomplete despatches as possible.

Suspend Processing of Stock

The movement of stock must be suspended whilst the stock count for Data Take-on is performed and reconciled.

Movement of stock can be resumed after the activation of the warehouse.

Initial Data Take-On / Reconciliation

A full stock count is required to include every item and location in the warehouse. The details can either be entered through the Data Take-On data entry programs or by recording Physical Stock Adjustments in Drinks Inventory.

If Data Take-On is used Reconciliation with Update will set the Drinks Inventory balances.

Resume Processing of Stock

The warehouse must be activated. This is done by taking the appropriate function key from the Warehouse Profile Maintenance option.

Transaction processing can be resumed in the warehouse.

Receipts and issues are now subject to the control of the Drinks Warehousing module.

Installing Application Manager PTF's

Application Manager PTF's must be restored and installed as QSECOFR.



Note: To install the Application Manager PTF's the system must be in a minimum state (that is with no users or subsystems / jobs running - this is important)

Use a normal User ID to create the state then sign off and sign on as QSECOFR.

At the first screen, add in library OSLPTP1 (ADDLIB OSLPTP1 *FIRST)

Type in MNGPTF and install the PTF as normal.

Once installed, the library and PTF object can only be removed by QSECOFR. Install this PTF then sign off and sign back in as PTFOwner for subsequent PTF's.

If the above is NOT done, PTFOwner, being inside Application manager, will say the AM PTF is installed when it has not, as PTFOwner is using one of the objects the PTF requires, and the system will work - but will NOT have been updated.